

Gate Smashers OS Playlist Exam Mapping Guide

Course Pattern: TE IT - 2024 Pattern (SPPU)
Document Type: Exam Playlist Mapping

Academic Term: Semester V (2026)
Course Code: PCC-301-ITT

Introduction and Purpose

This document provides an exam-focused, structured mapping of the Gate Smashers Operating System (OS) YouTube playlist to the official SPPU TE IT (2024 Pattern) Operating System Syllabus. It classifies all 92 videos from the playlist into the five core syllabus units.

Exam Preparation Note: Lectures highlighted in **bold** represent sure-shot exam topics that are explicitly tested in SPPU end-semester exams and fully covered in the unit-wise theory notes (such as System Calls, CPU Scheduling, Semaphores, Banker's Algorithm, Paging, TLB, and Disk Scheduling). Focus on these lectures first for maximum marks!

Quick Statistics:

- **Total Playlist Videos Extracted:** 92 (Total Duration: 20h 18m)
- **Unit 1 (Overview of OS):** 12 Videos (2h 6m)
- **Unit 2 (Process Description & Control):** 11 Videos (1h 51m)
- **Unit 3 (Concurrency Control & Deadlocks):** 21 Videos (6h 37m)
- **Unit 4 (Memory Management & Virtual Memory):** 27 Videos (6h 9m)
- **Unit 5 (I/O & File Management):** 21 Videos (3h 33m)

Unit 1: Overview of Operating System (2h 6m)

Syllabus Topics Covered: OS Objectives & Functions, Evolution of OS, Developments Leading to Modern OS, Virtual Machines, Dual-Mode Execution, System Calls, Fork(), and BASH Linux commands.

Lecture Topic / Title	Duration	Watch Link
L-1.1: Introduction to Operating System and its Functions with English Subtitles	18:59	Play ↗
L-1.2: Batch Operating System - Types of Operating System	7:12	Play ↗
L-1.3: Multiprogramming and Multitasking Operating System in Hindi with real life examples	6:34	Play ↗
L-1.4: Types of OS(Real Time OS, Distributed, Clustered & Embedded OS)	8:15	Play ↗
L-1.5: Process States in Operating System- Schedulers(Long term,Short term,Medium term)	20:54	Play ↗
L-1.6: Imp Linux Commands(Operating System) - Must Watch for College/University & Competitive exams	10:58	Play ↗
L-1.7: System Calls in Operating system and its types in Hindi	10:07	Play ↗
L-1.8: Fork System call with Example - Fork() system call questions	10:02	Play ↗

Lecture Topic / Title	Duration	Watch Link
L-1.9: Questions on Fork System Call With Explanation - Operating System	6:39	Play ↗
L-1.10: User mode and Kernel mode in operating system in hindi	6:46	Play ↗
L-1.11: Process Vs Threads in Operating System	11:17	Play ↗
L-1.12: User Level Vs Kernel Level Thread in Operating System - All Imp Points	8:29	Play ↗

Unit 2: Process Description and Control (1h 51m)

Syllabus Topics Covered: Process Concept, 5-state & 7-state models, PCB, ULT vs KLT, POSIX Pthreads, Preemptive vs Non-Preemptive CPU Scheduling (FCFS, SJF, SRTF, Round Robin, Priority, Multilevel Queue, MLFQ).

Lecture Topic / Title	Duration	Watch Link
L-2.1: Process Scheduling Algorithms (Preemption Vs Non-Preemption) - CPU Scheduling in OS	10:31	Play ↗
L-2.2: What is Arrival, Burst, Completion, Turnaround, Waiting and Response time in CPU Scheduling	8:59	Play ↗
L-2.3: First Come First Serve(FCFS) CPU Scheduling Algorithm with Example	10:34	Play ↗
L-2.4: Shortest Job First(SJF) Scheduling Algorithm with Example - Operating System	9:14	Play ↗
L-2.5: Shortest Remaining Time First (SJF With Preemption) Scheduling Algorithm with Example - OS	13:37	Play ↗
L-2.6: Question on Shortest Job First(SJF) with Preemption Scheduling Algorithm	7:52	Play ↗
L-2.7: Round Robin(RR) CPU Scheduling Algorithm with Example	16:23	Play ↗
L-2.8: Pre-emptive Priority Scheduling Algorithm with Example - Operating System	10:30	Play ↗
L-2.9: Example of Mix Burst Time(CPU & I/O both) in CPU Scheduling - Tough Question	14:19	Play ↗
L-2.10: Multi Level Queue Scheduling - Operating System	3:41	Play ↗
L-2.11: Multilevel Feedback Queue Scheduling - Operating System	5:22	Play ↗

Unit 3: Concurrency Control (6h 37m)

Syllabus Topics Covered: Process Synchronization, Race Condition, Critical Section, Test & Set, Turn Variable, Counting & Binary Semaphores, Producer-Consumer, Readers-Writers, Dining Philosophers, Resource Allocation Graphs, Deadlock Prevention, Banker's Algorithm.

Lecture Topic / Title	Duration	Watch Link
L-3.1: Process Synchronization - Process Types - Race Condition - Operating System-1	17:02	Play ↗
L-3.2: Producer Consumer Problem - Process Synchronization Problem in Operating System	26:36	Play ↗
L-3.3: Printer-Spooler Problem - Process Synchronization Problem in Operating System	13:15	Play ↗
L-3.4: Critical Section Problem - Mutual Exclusion, Progress and Bounded Waiting - Operating System	25:36	Play ↗
L-3.5: LOCK Variable in OS - Process Synchronization	7:24	Play ↗
L-3.6: Test and Set Instruction in OS - Process Synchronization	9:09	Play ↗
L-3.7: Turn Variable - Strict Alteration Method - Process Synchronization	8:21	Play ↗
L-3.8: Semaphores - Wait, Signal Operation - Counting Semaphore - Example- Operating system	24:42	Play ↗
L-3.9: What is Binary Semaphore - Easiest Explanation - Operating system	12:23	Play ↗
L-3.10: Practice Question on Binary Semaphore in Operating System	16:00	Play ↗
L-3.11: Solution of Producer Consumer Problem using Semaphore - Operating System	17:30	Play ↗
L-3.12: Solution of Readers-writers Problem using Binary semaphore	21:11	Play ↗
L-3.13: Dining philosophers Problem and Solution using Semaphore in Operating System	35:13	Play ↗
L-4.1: DEADLOCK concept - Example - Necessary condition - Operating System	12:21	Play ↗
L-4.2: Resource Allocation Graph in Deadlock - Single Instance with example - Operating System	26:18	Play ↗
L-4.3: Multi-Instance Resource Allocation Graph with Example - Operating System	21:42	Play ↗
L-4.4: Deadlock Handling Methods and Deadlock Prevention - Operating System	24:49	Play ↗
L-4.5: Deadlock Avoidance Banker's Algorithm with Example -With English Subtitles	24:04	Play ↗
L-4.6: GATE 2018 Question on Banker's Algorithm - Deadlock avoidance - Operating System	16:02	Play ↗
L-4.7: Question Explanation on Deadlock - Operating System	15:46	Play ↗
L-4.8: GATE 2018 Question Explanation on deadlock - Operating system	22:15	Play ↗

Unit 4: Memory Management (6h 9m)

Syllabus Topics Covered: Contiguous & Non-contiguous allocation, Fixed/Variable partitioning, First/Best/Worst fit, Paging, Page tables, 2-Level & Inverted Paging, Segmentation, Virtual Memory, Demand Paging, TLB, Page replacement (FIFO, Optimal, LRU, MRU), Belady's Anomaly, Thrashing.

Lecture Topic / Title	Duration	Watch Link
L-5.1: Memory Management and Degree of Multiprogramming - Operating System	23:41	Play ↗
L-5.2: Memory management Techniques - Contiguous and non-Contiguous - Operating System	11:46	Play ↗
L-5.3: Internal Fragmentation - Fixed size Partitioning - Memory management - Operating System	18:53	Play ↗
L-5.4: Variable size Partitioning - Memory management - Operating System	15:11	Play ↗
L-5.5: First Fit, Next Fit, Best Fit, Worst fit Memory Allocation - Memory Management - OS	16:22	Play ↗
L-5.6: GATE Question Solved on First Fit, Best Fit and Worst fit Memory Allocation - Operating System	14:02	Play ↗
L-5.7: GATE 2007 Question Solved on First Fit, Best Fit and Worst fit with timeline - OS	15:41	Play ↗
L-5.8: Need of Paging - Memory Management - Operating System	14:20	Play ↗
L-5.9: What is Paging - Memory management - Operating System	25:54	Play ↗
L-5.10: Question Explanation on Logical address and Physical address space - Operating System	15:25	Play ↗
L-5.11: Question Explanation on Paging - Memory Management - Operating System	13:52	Play ↗
L-5.12: Page Table Entries - Format of Page Table - Operating System	10:39	Play ↗
L-5.13: 2-Level Paging in Operating System - Multilevel Paging	15:45	Play ↗
L-5.14: Inverted paging - Memory Management - Operating System	10:35	Play ↗
L-5.15: Paging Questions in Operating System(OS) - Imp Question for all competitive exams	7:53	Play ↗
L-5.16: What is Thrashing - Operating System	8:34	Play ↗
L-5.17: Segmentation Vs Paging - Segmentation Working - Operating system	16:30	Play ↗
L-5.18: Overlay - Memory Management - Operating system	11:12	Play ↗
L-5.19: Virtual Memory - Page fault - Significance of virtual memory - Operating System	20:18	Play ↗
L-5.20: Translation Lookaside Buffer(TLB) in Operating System in Hindi	12:23	Play ↗
L-5.21: Numerical on Translation Lookaside Buffer (TLB) - Operating System	4:31	Play ↗

Lecture Topic / Title	Duration	Watch Link
L-5.22: Page Replacement Introduction - FIFO Page Replacement algorithm - Operating System	16:07	Play ↗
L-5.23: Belady's Anomaly in FIFO page Replacement with example - Operating System	12:26	Play ↗
L-5.24: Optimal Page Replacement algorithm - Operating System	13:07	Play ↗
L-5.25: Least Recently Used Page Replacement Algorithm - Operating System	7:57	Play ↗
L-5.26: Most recently used page replacement algorithm - Operating System	6:42	Play ↗
TLBs Toughest Question asked in GATE exam - Operating System	9:53	Play ↗

Unit 5: Input/Output and File Management (3h 33m)

Syllabus Topics Covered: Hard Disk Architecture, Disk Access Time, Disk Scheduling (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK), File System overview, File attributes & operations, Contiguous, Linked & Indexed file allocation, Unix Inode Structure, Protection & Security.

Lecture Topic / Title	Duration	Watch Link
Lec-0: Operating System Syllabus Discussion for all College/University & Competitive exams (GATE, NET)	13:01	Play ↗
L-6.1: Hard Disk Architecture in Operating System in Hindi	11:35	Play ↗
L-6.2: Disk Access Time with Example - Seek Time, Rotational Time and Transfer Time	15:13	Play ↗
L-6.3: Disk Scheduling Algorithm - Operating System	6:08	Play ↗
L-6.4: FCFS in Disk scheduling with Example - Operating System	8:44	Play ↗
L-6.5: SSTF in Disk scheduling with Example - Operating System	8:30	Play ↗
L-6.6: SCAN Algorithm in Disk scheduling with Example - Operating System	7:54	Play ↗
L-6.7: LOOK Algorithm in Disk scheduling with Example - Operating System	4:28	Play ↗
L-6.8: C-SCAN Algorithm in Disk scheduling with Example - Operating System	5:22	Play ↗
L-6.9: C-LOOK Algorithm in Disk scheduling with Example - Operating System	4:59	Play ↗
L-6.10: Important Questions on Operating System - Must Watch - NTA NET June 2021	9:41	Play ↗
L-7.1: File System in Operating System - Windows, Linux, Unix, Android etc.	9:55	Play ↗
L-7.2: File Attributes & Operations in Operating System	11:14	Play ↗

Lecture Topic / Title	Duration	Watch Link
L-7.3: Allocation Methods in operating system in hindi - Contiguous vs NonContiguous	5:59	Play ↗
L-7.4: Contiguous Allocation in Operating system - Advantages & Disadvantages	12:37	Play ↗
L-7.5: Linked List allocation in file allocation with example - Operating system	10:55	Play ↗
L-7.6: Indexed File Allocation in Operating System	10:43	Play ↗
L-7.7: Unix Inode Structure with Numerical Example - OS	10:26	Play ↗
Lec-8: Protection & Security in Operating system - Full OS playlist	20:54	Play ↗
Top 15 OS Interview Questions - Operating System Interview - Placement Strategy	15:37	Play ↗
Linker & Loader with example	9:54	Play ↗

End of Operating Systems Gate Smashers Playlist Mapping — SPPU TE IT 2024 Pattern